

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BITTOE604E - INTRODUCTION TO DATA SCIENCE

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which phase of the data science process uses descriptive statistics and visual techniques?
 1. Model Building
 2. Data Exploration
 3. Data Collection
 4. Data Presentation
- 2 In the data science process, what is the primary goal of the 'Data Preparation' phase?
 1. Building machine learning models
 2. Presenting final results
 3. Creating visualizations
 4. Removing errors and integrating data
- 3 What will be the output of the following code?

```
x = "2"  
y = 3  
print(x * y)
```

 1. 6
 2. "6"
 3. "222"
 4. Error
- 4 What is the result of the dot product of vectors [1, 2] and [3, 4]?
 1. 10
 2. 11
 3. 12
 4. 13

- 5 Which distribution describes the number of successes in a fixed number of trials?
1. Normal
 2. Exponential
 3. Poisson
 4. Binomial
- 6 In hypothesis testing, the p-value represents:
1. The probability that the null hypothesis is true
 2. The effect size
 3. The probability of observing the data if the null hypothesis is true
 4. The confidence level
- 7 In binary logistic regression, the dependent variable is:
1. Continuous
 2. Categorical with 3 or more classes
 3. Ordinal
 4. Binary (0 or 1)
- 8 Gain in a gain chart is defined as:
1. Number of false positives
 2. Cumulative % of positives captured
 3. Ratio of true negatives
 4. Difference in predicted vs actual
- 9 A model predicts house prices using the number of bedrooms, area in sqft, and location. You want to know how much each factor affects price. Which method will best help you analyze this?
1. K-Means Clustering
 2. Simple Linear Regression
 3. Multiple Linear Regression
 4. Logistic Regression
- 10 What is the process of converting raw data into a structured format called?
1. Data Visualization
 2. Data Munging (Wrangling)
 3. Web Scraping
 4. Model Training
- 11 Which technique is used to reduce the number of features in a dataset while preserving important information?
1. Data Scaling
 2. One-Hot Encoding
 3. Dimensionality Reduction
 4. Web Scraping
- 12 Scikit-learn is mainly used for:
1. Web scraping
 2. Machine learning algorithms
 3. Natural language processing
 4. Deep learning

Q2. Solve the following.

A) Explain data science process along with detailed diagram. 6

B) List and explain the Facets/Types of Data in Data Science. 6

Q3. Solve the following.

A) Discuss the following central tendencies: mean, median and mode. Find mean, median and mode for the list List1 = [8, 16, 13, 11, 15, 9, 12, 14]. 6

B) Consider a random experiment of tossing a biased coin 6 times where the probability of getting a head is 0.6. If 'getting a head' is considered as 'success' then, the binomial distribution table will contain the probability of r successes for each possible value of r . Write down python code to find out Mean and Variance. Also write down code to plot the graph using matplotlib() in vertical bars 6

r	0	1	2	3	4	5	6
P(r)	0.004096	0.036864	0.138240	0.276480	0.311040	0.186624	0.046656

Q4. Solve Any Two of the following.

A) Discuss different machine learning algorithms that can be applied to weather 6 forecasting tasks. Compare models like Decision Trees, Random Forests, and Support Vector Machines (SVM) in terms of their suitability for this application.

B) Write a short on Decision Tree classifier. 6

C) Write simple Python code for building the Linear Regression Model. (Assume suitable 6 data).

Q5. Solve Any Two of the following.

A) Demonstrate various approaches to deal with missing values. Fill the missing values by 6 using mean of that column.

	a	b	c	d	e
0	23	49.0	47.0	15.0	16
1	22	NaN	27.0	48.0	38
2	46	47.0	NaN	10.0	24
3	36	20.0	NaN	31.0	24
4	12	49.0	NaN	NaN	39
5	49	15.0	43.0	33.0	16
6	15	21.0	43.0	NaN	39
7	15	43.0	17.0	39.0	25
8	39	18.0	30.0	16.0	21
9	16	27.0	30.0	26.0	44

B) Write a note on Dimensionality Reduction Techniques. 6

C) Write a Python program to scrape press release titles and dates from a government health 6 agency website (e.g., FDA, NIH, or CDC) that mention "AI" or "artificial intelligence".

- Filter and extract only the relevant press releases.
- Save the data in a CSV file.
- Generate a bar chart showing the number of such press releases per year.

Q6. Solve Any Two of the following.

- A) Write a Python program to build a simple recommendation system using collaborative filtering. You may use a sample dataset, such as MovieLens, for this task. Describe the steps involved in loading the dataset, pre-processing the data, and applying the collaborative filtering technique.
- B) Describe how you would formulate the detection of malicious URLs as a multi-class classification problem. What would be the different classes, and how would you label the dataset?
- C) A company has recorded its monthly sales data over the past 5 years. Discuss the components of a Time Series and how they influence forecasting. Using this dataset, explain how you would apply a Moving Average for smoothing, perform Decomposition to isolate trend and seasonality, and build and evaluate an ARIMA model for forecasting future sales.

*** End ***